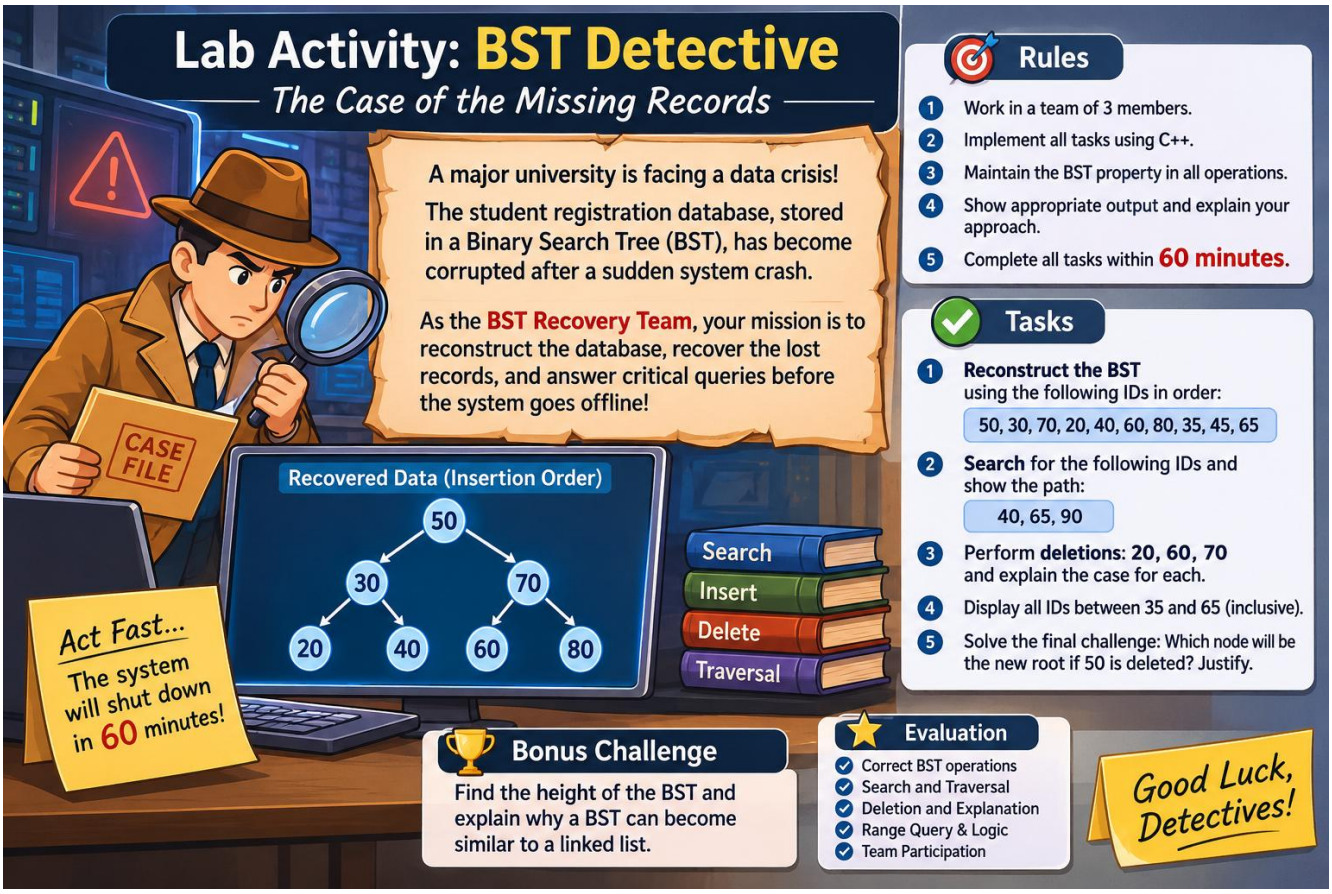
BST DETECTIVE

The Case of the Missing Records

BINARY SEARCH TREE – SCENARIO-BASED LAB ACTIVITY



1. Scenario: The University Database Crisis

A major university stores student registration records in a Binary Search Tree (BST). Following a sudden system crash, the database has become inaccessible. The recovery team has retrieved the registration IDs, but the original database structure is missing.

Your team has been appointed as the BST Recovery Team. Reconstruct the database, investigate records, remove invalid entries, and answer urgent queries before the system shuts down.

2. Rules of the Mission

1. Work in a team of three. Assign the roles of Database Architect, Detective, and System Engineer.
2. Implement the required operations using C++.
3. For every insertion, follow the BST ordering property: values smaller than a node go to its left; larger values go to its right.
4. Do not rearrange the given insertion sequence. Insert the IDs in exactly the order provided.
5. After each deletion, display or draw the updated tree and explain how the BST property is preserved.
6. Show the output for every task. A correct answer without a trace or explanation may not receive full marks.
7. Complete all missions within 60 minutes. You may use paper to sketch the tree before coding.

3. Mission Tasks

Mission 1 – Reconstruct the Database

Insert the following student registration IDs into an initially empty BST, in the exact order shown:

50, 30, 70, 20, 40, 60, 80, 35, 45, 65

- Draw the final BST clearly, showing every node and its left and right child.
- Identify the root node, all leaf nodes, and the internal nodes.
- Find the minimum and maximum registration IDs.

Mission 2 – Investigate the Missing Records

Search for each ID below in the BST. For every search, write the sequence of nodes visited from the root and state whether the ID is found.

Search keys: 40, 65, 90

- Example format: Search key → nodes visited → Found / Not Found.

Mission 3 – Clean the Database

The following records have been declared invalid. Delete them one after another from the BST created in Mission 1:

Delete: 20 → 60 → 70

- After each deletion, draw or display the updated BST.
- For each deleted node, identify the deletion case: leaf node, node with one child, or node with two children.
- If a node has two children, explain whether you used its inorder successor or inorder predecessor.

Mission 4 – Recover Records Within a Range

The university needs a list of all registration IDs from 35 to 65, inclusive. Also mention which part of subtree you will get more data. Explain how the BST property can help your function avoid exploring irrelevant parts of the tree.

Mission 5 – Final Recovery Challenge

The administrator asks: “If the original BST is reconstructed and the node 50 is deleted, which value becomes the root of the tree?”

Perform the deletion on a fresh copy of the original BST, show the resulting tree, and justify your answer using the deletion rule for a node with two children.

********** ***Mission Complete? Submit Your Code, Tree Diagrams, And Explanations*** **********